12.82

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Question:
let $\mathbf{M}$ be a 3× 3 symmetric matrix with eigenvalues 1,2 and 3. let $\mathbf{N}$ be any 3× 3 matrix with real eigenvalues such that $\mathbf{MN}+\mathbf{NM} = 3\mathbf{I}$, where $\mathbf{I}$ is a 3× 3 identity matrix. Then which of the following cannot be eigenvalue(s) of the matrix $\mathbf{N}$? (ST 2022)

a) $\frac{1}{4}$ b) $\frac{3}{4}$ c) $\frac{1}{2}$ d) $\frac{7}{4}$

Solution:
let λ_1 , λ_2 , λ_3 be the eigenvalues of $\mathbf{N}$
By Eigenvalue decomposition

$$\mathbf{M} = \mathbf{QDQ}^\top \tag{1}$$

where

$$\mathbf{Q}^\top\mathbf{Q} = \mathbf{QQ}^\top = \mathbf{I} \tag{2}$$
$$\mathbf{D} = \text{diag}\,(1, 2, 3) \tag{3}$$
$$\implies \mathbf{D} = \mathbf{Q}^\top\mathbf{MQ} \tag{4}$$

$$\mathbf{MN} + \mathbf{NM} = 3\mathbf{I} \tag{5}$$
$$\mathbf{Q}^\top\left(\mathbf{MN} + \mathbf{NM}\right)\mathbf{Q} = \mathbf{Q}^\top 3\mathbf{IQ} \tag{6}$$
$$\mathbf{Q}^\top\mathbf{MNQ} + \mathbf{Q}^\top\mathbf{NMQ} = 3\mathbf{I} \tag{7}$$
$$\implies \mathbf{Q}^\top\mathbf{MNQ} = \left(\mathbf{Q}^\top\mathbf{MQ}\right)\left(\mathbf{Q}^\top\mathbf{NQ}\right) \tag{8}$$
$$\implies \mathbf{Q}^\top\mathbf{NMQ} = \left(\mathbf{Q}^\top\mathbf{NQ}\right)\left(\mathbf{Q}^\top\mathbf{MQ}\right) \tag{9}$$

let

$$\mathbf{Q}^\top\mathbf{NQ} = \mathbf{N}' \tag{10}$$
$$\implies \mathbf{N} = \mathbf{QN}'\mathbf{Q}^\top \tag{11}$$
$$\implies \mathbf{Q}^\top\mathbf{MNQ} = \mathbf{DN}' \tag{12}$$
$$\implies \mathbf{Q}^\top\mathbf{NMQ} = \mathbf{N}'\mathbf{D} \tag{13}$$

From (11) and (12) (7) can be written as

$$\mathbf{DN}' + \mathbf{N}'\mathbf{D} = 3\mathbf{I} \tag{14}$$

let $(\mathbf{DN}')_{i,j}$ be the entry of ith column and jth row of the matrix $\mathbf{DN}'$. As $\mathbf{D}$ is a diagonal matrix

$$(\mathbf{DN}')_{i,j} = \mathbf{D}_i\mathbf{N}'_{i,j} \tag{15}$$

let $(\mathbf{N}'\mathbf{D})_{i,j}$ be the entry of ith column and jth row of the matrix $\mathbf{N}'\mathbf{D}$. As $\mathbf{D}$ is a diagonal matrix

$$(\mathbf{N}'\mathbf{D})_{i,j} = \mathbf{D}_j\mathbf{N}'_{i,j} \tag{16}$$
$$\implies \left(\mathbf{D}_i + \mathbf{D}_j\right)\mathbf{N}'_{i,j} = 3\,(\mathbf{I})_{i,j} \tag{17}$$

for $i \neq j$

$$\left(\mathbf{D}_i + \mathbf{D}_j\right)\mathbf{N}'_{i,j} = 0 \tag{18}$$
$$\implies \mathbf{N}'_{i,j} = 0 \tag{19}$$

for $i = j$

$$\left(\mathbf{D}_i + \mathbf{D}_j\right)\mathbf{N}'_{i,j} = 3 \tag{20}$$
$$\implies \mathbf{N}'_{i,j} = \frac{3}{2D_i} \tag{21}$$

$\implies$ The diagonal values of $\mathbf{N}'$ are $\frac{3}{2}$ and $\frac{3}{4}$ and $\frac{1}{2}$
In (11) $\mathbf{Q}$ is an orthogonal matrix and $\mathbf{N}'$ is a daigonal matrix so we can say that (11) is the eigenvalue decomposition of $\mathbf{N}$
$\implies \mathbf{N} = \mathbf{QN}'\mathbf{Q}^\top$

$$\mathbf{N}' = \text{diag}\,(\lambda_1, \lambda_2, \lambda_3) \tag{22}$$
$$\implies \lambda_1 = \frac{3}{2} \text{ and } \lambda_2 = \frac{3}{4} \text{ and } \lambda_3 = \frac{1}{2} \tag{23}$$